

IBM HR ANALYTICS PROJECT – DOCUMENTATION

1. PROJECT OVERVIEW



The objective of this project is to analyze employee data to understand workforce distribution and identify the key factors contributing to employee attrition. The insights will help HR teams to improve employee retention and workforce planning.

2. DATASET INFORMATION

Dataset Name	: IBM HR Analytics Employee Attrition
Source	: IBM Sample Data
Total Rows	: 1,470
Total Columns	: 35
File Format	: CSV
Description	: Contains employee demographic, job role, performance, salary and attrition related information.

3. TOOLS AND TECHNOLOGIES USED

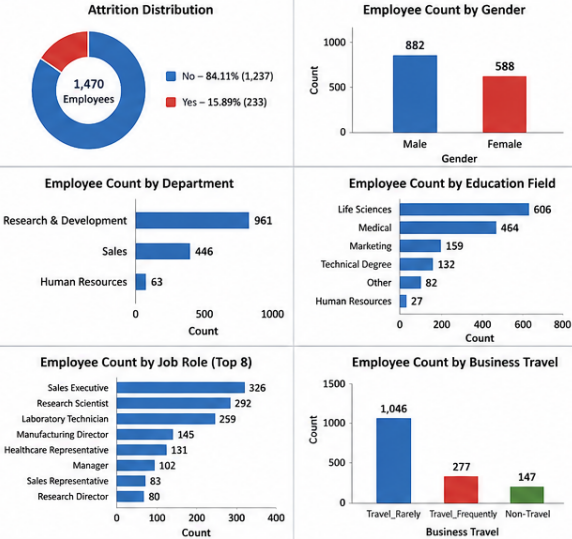


4. DATA CLEANING AND PREPROCESSING

- ✓ Checked and removed duplicate records.
- ✓ Handled missing values.
- ✓ Corrected data types.
- ✓ Standardized categorical values.
- ✓ Created new columns where required.
- ✓ Verified data consistency and accuracy.



5. EXPLORATORY DATA ANALYSIS (EDA) – KEY VISUALS



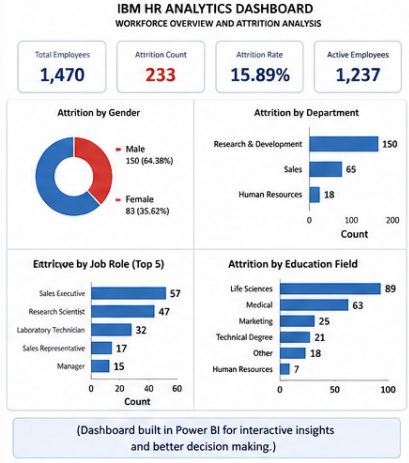
6. KEY INSIGHTS

- 15.89% of employees have left the organization.
- Most of the employees are from Research & Development department.
- Life Sciences background employees are the highest in number.
- Sales Executive and Research Scientist are the top job roles.
- Employees who travel rarely are the majority.

7. RECOMMENDATIONS

- Focus on retention strategies for at-risk employees.
- Improve work-life balance and job satisfaction.
- Provide growth and learning opportunities.
- Recognize and reward high performers.
- Monitor attrition trends regularly.
- Strengthen employee engagement programs.

8. POWER BI DASHBOARD



9. CONCLUSION



This analysis provides valuable insights into employee attrition and workforce distribution. By understanding the key factors, the organization can take proactive steps to improve employee satisfaction and reduce attrition rate.

10. FUTURE IMPROVEMENTS

- Build predictive model to identify potential attrition.
- Include more employee feedback data.
- Automate reports and dashboard refresh.
- Integrate with HR systems for real-time analytics.



PROJECT BY:
[Your Name]



DATE:
May 12, 2024



ENVIRONMENT:
Jupyter Notebook, Power BI Desktop



DATA SOURCE:
IBM Sample Employee Data



CONTACT:
your.email@example.com